Finding the Nuclei That Models Miss

Reliable instance segmentation and classification
across diverse tissues in PanNuke

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Project 5

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Project summary

Missed nuclei distort measurements of tissue composition, even when a segmentation model achieves a strong average score. Rare, small, and partially visible cells are particularly vulnerable; in our PanNuke experiments, the largest detection deficit affects the Dead class, which includes apoptotic and necrotic nuclei.

We will investigate this deficit through two complementary approaches. A controlled supervision study tests whether retaining geometric targets for cropped nuclei improves detection at image boundaries. PM-Net, our proposed architecture, predicts individual masks and cell types from point-conditioned queries, replacing horizontal/vertical distance maps and watershed decoding. Comparisons under a matched training protocol will test whether direct instance prediction improves detection while maintaining segmentation quality and practical inference cost.

Intellectual merit

This study tests how training-target construction and instance representation affect sensitivity to rare and partially visible nuclei. A shared encoder and evaluation protocol, combined with separate supervision and decoder ablations, will help identify which design choices account for changes in detection and mask quality.

Broader impacts

Reliable nucleus counts and cell labels support quantitative studies of tissue composition and the tumor microenvironment. Shared evaluation code and tissue-specific error analyses will make improvements in rare-class detection easier to assess alongside segmentation quality and computational cost.

1. Research aims, context, and preliminary evidence

Nuclei in hematoxylin and eosin (H&E) images vary in staining, morphology, and tissue context, while close neighbors can be difficult to separate. Missed apoptotic or necrotic nuclei, merged instances, and type confusion can bias estimates of cell composition. With instance masks and five cell classes across 19 tissues, PanNuke supports comparisons of these errors across diverse tissue environments [1].

Specific aims

Aim 1 - Characterize segmentation and typing failures. Consolidate HoVer-Net and CellViT-UNI results under a shared evaluator, separating missed nuclei, merges, splits, and type errors by tissue and class.

Aim 2 - Evaluate supervision for truncated nuclei. Complete the four-arm, two-seed comparison of stronger cropping and selective HV supervision, using both fixed crop windows and original validation images.

Aim 3 - Develop a point-conditioned instance model. Test whether PM-Net improves Dead detection and crowded-instance separation within predefined quality and inference-cost limits.

Prior work and architectural motivation

HoVer-Net separates instances using horizontal/vertical (HV) offset maps [2], while CellViT combines a transformer encoder with nuclei segmentation and classification [3]. Our reproduced CellViT-UNI provides the primary baseline. PM-Net brings together point-conditioned masks from PromptNucSeg [4], query-based prediction from Mask2Former [5], and spatial adaptation from CellVTA [6]. Both models will use the same pretrained UNI encoder [10] and unfreezing schedule, with adapter controls to distinguish decoder effects from added capacity.

Data and baseline evidence

PanNuke contains 7,901 RGB patches at 256 x 256 pixels and roughly 200,000 annotated nuclei. Its three prepared folds contain 2,656, 2,523, and 2,722 patches. Each instance is labeled neoplastic, inflammatory, connective, dead, or epithelial. Development uses fold 1 for training and fold 2 for validation.

Reproduced baselines: three-split means	mPQ	bPQ
HoVer-Net, final checkpoint, no test-time augmentation	0.4564	0.6613
CellViT-UNI, final checkpoint, no test-time augmentation	0.4995	0.6653

Earlier analysis found that 38.5% of labeled Dead nuclei had intersection-over-union (IoU) below 0.1 with every predicted object. Tested reclassification, augmentation, and frozen-feature probes provided no consistent improvement [7]. Combining predictions from two scales improved Dead PQ modestly at about 6.1 times the inference cost, motivating a focus on instance detection and partial visibility within a single model.

2. Aim 2: supervision for truncated nuclei

Spatial cropping can leave only part of a nucleus visible at a training-image boundary. The existing target generator sets HV values to zero for instances smaller than 30 pixels. Aim 2 tests whether retaining HV supervision for newly truncated fragments improves detection, using identical images and other labels across the matched intervention arms.

Arm	Crop probability	HV target rule
A0: retrained baseline	0.1	Minimum area 30
A1: crop-only control	0.5	Minimum area 30
A2: selective supervision	0.5	Minimum 1 for newly truncated instances with visible area below 30
A3: small-object control	0.5	Minimum 1 for all instances; targets differ only below area 30

All arms use seeds 19 and 1, 130 epochs, matched initialization and optimization, and final-checkpoint evaluation at native resolution without test-time augmentation (TTA). A1-A3 share the augmentation random sequence. Instance identities before and after cropping distinguish newly truncated objects from those already cut by a patch boundary; areas are measured after geometric transformations [8].

Interim findings

An interim development snapshot comprises six completed fold-2 evaluations: all four arms for seed 19, and A0/A1 for seed 1. Relative to A0, A2 seed 19 raises Dead PQ by 0.0222 and strict Dead PQ by 0.0025, but lowers mPQ by 0.0091 and bPQ by 0.0062. The overall losses exceed the predefined 0.002 tolerance, indicating a tradeoff that fails the quality criterion for expansion [8].

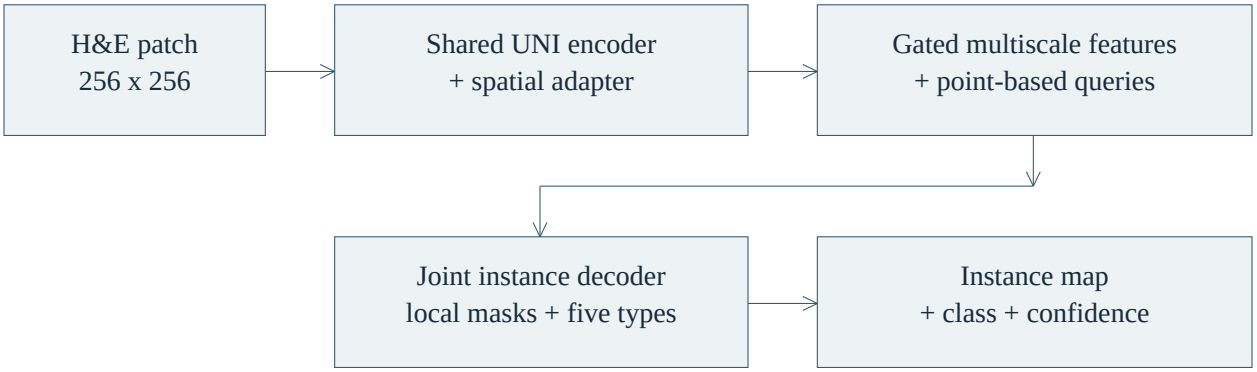
Mechanism analysis

The remaining analysis uses 2,523 fixed crop windows to measure misses among newly truncated Dead nuclei. A2-A1 isolates selective supervision, A2-A0 quantifies its overall tradeoff, and A3 tests whether the effect extends to all small instances. The frozen rule requires fewer misses in both seeds, a mean reduction of at least 0.03, and a positive 95% lower confidence bound. Additional Dead and strict-score conditions remain as specified in the original protocol [8].

Aim 3 tests a complementary hypothesis: predicting each instance mask directly may improve detection of partial or crowded nuclei. Where Aim 2 changes supervision within the existing HV decoder, PM-Net changes the instance representation under a matched encoder and training schedule. Together, the two studies separate the effects of target construction from those of decoder design.

3. Aim 3: PM-Net, a direct instance model

PM-Net will predict a set of nucleus masks and cell types conditioned on candidate locations [9]. UNI provides tissue context, supplemented by a convolutional adapter for fine spatial features in small and crowded nuclei. The prototype will use native 256 x 256 patches and the same pretrained encoder initialization as CellViT-UNI [10].



Proposed PM-Net. One shared image encoding; no HV maps or watershed.

Architecture and optimization

Shared image features. A compact CNN adapter supplements UNI features with higher-resolution spatial information. Learned gates combine the feature scales before the point and mask heads.

Point-conditioned queries. A supervised heatmap predicts candidate nucleus locations and types. Each selected point initializes an instance query, a feature vector for one proposed nucleus. Training targets lie inside visible instance masks. Query capacity, initially about 256, will be set from training-set object counts with a safety margin; overflow will be reported.

Joint mask and type prediction. Queries attend to local features and tissue context to predict masks, confidence, and five-class probabilities. One-to-one matching pairs predictions with labeled nuclei using location, class, and mask agreement. Training combines point focal loss, mask cross-entropy and Dice losses, and type cross-entropy, with weights selected on the validation fold. The point head receives its own supervised loss because discrete point selection does not pass gradients.

Instance decoding. A validation-selected confidence rule resolves overlapping masks, which are mapped to native image coordinates for scoring. A separate analysis will measure how minimum-area filtering affects small-instance recall, distinguishing post-processing effects from the training-target cutoff studied in Aim 2.

Controlled ablations

A basic query decoder will be compared with versions that add point initialization and gated fusion separately. A plain adapter with matched parameter count will distinguish architectural effects from added capacity. The initial study retains five-class prediction throughout; truncation supervision and auxiliary type grouping are subsequent extensions rather than prerequisites for the core model.

4. Experimental design and evaluation

Development and confirmation

Development pairs seeds 19 and 1 on training fold 1 and validation fold 2, using the existing 130-epoch schedule and encoder unfreezing at epoch 25. The encoder initialization and training recipe will be matched across PM-Net and CellViT-UNI. Architecture, losses, augmentation, decoding, and decision criteria will be fixed before confirmation.

If the prototype meets the development criteria and resources permit, confirmation will follow the project train/validation/test protocol: 1/2/3, 2/1/3, and 3/2/1. Final checkpoints will be evaluated for every split and seed before averaging. Splits 1 and 2 share test fold 3. Earlier stages have examined the test folds, and available identifiers cannot establish patient-level separation; both limitations will accompany the results. Optional external evaluation on CoNIC and MoNuSAC will use class mapping and overlap exclusion.

Endpoints and statistical analysis

The shared PanNuke evaluator measures official mPQ, which includes type, bPQ, which ignores type, and per-class panoptic quality (PQ). Strict mPQ and strict Dead PQ additionally retain penalties for predictions of classes absent from an image. Detection and typing errors will be separated, with Dead recall stratified by tissue and by interior versus border location. Merge/split rates and tissue-by-class heatmaps will characterize crowded-instance performance.

Comparisons with matched CellViT-UNI and the crop-only control will use paired, tissue-stratified image bootstrap intervals. Crop windows from each source patch will form a single resampling cluster. Each seed will be reported separately, since these intervals measure image-sampling variation for the trained models. Class counts and annotation uncertainty will accompany rare-class estimates.

PM-Net development criteria

Before training, we propose fixing a primary target of at least 0.03 mean improvement in interior Dead recall, with a positive paired 95% lower confidence bound. Mean Dead PQ must improve by at least 0.005, and both recall and Dead PQ changes must be positive in each seed. Per-seed mPQ/bPQ losses are limited to 0.002; strict mPQ and strict Dead PQ must not decline, and mean merge/split rates must not increase. These criteria apply to PM-Net, independently of the frozen supervision experiment.

Inference, including decoding, must take no more than twice the matched single-scale baseline latency. Repeated measurements at the same hardware and batch settings will report latency, peak memory, and variation under shared-device load. Models meeting both quality and cost criteria will qualify for cross-split confirmation as resources permit.

5. Milestones, feasibility, and deliverables

Schedule	Milestone
Week 1	Complete the supervision analysis and baseline/error tables (minimum result).
Week 2	Finalize PM-Net design; implement and verify matching, masks, query capacity, and runtime.
Weeks 3-4	Complete the two-seed development comparison and core ablations (target result).
Weeks 5-6	Finalize the report; pursue cross-split and external evaluation if the prototype qualifies and resources permit.

The six-week schedule begins at proposal agreement, with three levels of scope:

Minimum result. Reproduced baselines, the completed four-arm supervision comparison, and a tissue- and class-specific error analysis with reproducible tables.

Target result. A tested PM-Net prototype and two-seed ablations that establish whether direct instance prediction improves Dead recall and PQ within the quality and latency criteria. A clear account of any remaining tradeoff is also a useful outcome.

Stretch goals. Cross-split confirmation of a qualifying prototype, external evaluation on CoNIC and MoNuSAC, and an exploratory auxiliary-supervision ablation. If the prototype falls short, Weeks 5-6 will focus on failure analysis and the final report.

Feasibility and main risks

Rare labels and annotation uncertainty. Tissue-specific counts and paired estimates will accompany rare-class results. Existing labels will be preserved, and disagreements among detectors will be retained as uncertain cases for future expert review.

Localization versus mask quality. Correctly located proposals can still yield masks below the matching threshold in crowded regions. Decomposing errors into missed points, insufficient mask overlap, duplicates, merges, and splits will identify the stage that limits performance.

Experimental scope and resources. The encoder and training recipe remain fixed during the decoder ablations. We estimate 2-4 A100 GPU-days for development and a further 4-8 for confirmation, subject to allocation. Available storage and checkpoint requirements will be checked before scheduling; resumable training will reduce losses from interruptions.

Deliverables and broader impact

The target deliverables are PM-Net code and trained models, versioned evaluation configurations, model/data provenance, comparison tables, and a final report with annotated failure cases. Together, these outputs will document how supervision and instance representation affect rare-nucleus detection under matched conditions.

This work supports more reliable measurements of cell composition across tissues. Reporting tissue-specific errors alongside aggregate scores will help researchers assess whether apparent improvements extend to underrepresented cells and justify their computational cost.

References and project evidence

[1] Gamper, J., et al. (2020). PanNuke Dataset Extension, Insights and Baselines.
https://arxiv.org/abs/2003.10778

[2] Graham, S., et al. (2019). HoVer-Net: Simultaneous segmentation and classification of nuclei in multi-tissue histology images. Medical Image Analysis, 58, 101563.
https://doi.org/10.1016/j.media.2019.101563

[3] Hörst, F., et al. CellViT: Vision Transformers for Precise Cell Segmentation and Classification. arXiv:2306.15350.
https://arxiv.org/abs/2306.15350

[4] Shui, Z., et al. Unleashing the Power of Prompt-driven Nucleus Instance Segmentation. arXiv:2311.15939.
https://arxiv.org/abs/2311.15939

[5] Cheng, B., et al. (2022). Masked-attention Mask Transformer for Universal Image Segmentation. CVPR.
https://arxiv.org/abs/2112.01527

[6] Yang, Y., et al. (2025). CellVTA: Enhancing Vision Foundation Models for Accurate Cell Segmentation and Classification. arXiv:2504.00784.
https://arxiv.org/abs/2504.00784

[7] BDD26 Project Team. (2026). PanNuke nuclei segmentation: research plan, experiment findings, and stage reports I-V. Internal project documentation, Johns Hopkins University.

Versioned project documentation

[8] BDD26 Project Team. (2026). Visibility-supervision experiment: frozen specification and interim development evaluations. Internal research protocol and experiment records, Johns Hopkins University.

Frozen supervision specification

[9] BDD26 Project Team. (2026). Next-stage research proposal: a new non-HoVer-Net model route. Internal design proposal, Johns Hopkins University.

Versioned new-model concept

[10] Chen, R. J., et al. (2024). Towards a general-purpose foundation model for computational pathology. *Nature Medicine*, 30, 850-862.

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Team contributions to this proposal

This page is separate from the main proposal.

The team will complete the statements below before submission.

TODO - COMPLETE BEFORE COURSE SUBMISSION
Add each team member's actual contribution to this proposal.

This section is intentionally left for the team to complete. Record work on the proposal itself, such as developing the research question, reviewing literature, designing experiments, preparing figures, or drafting and revising sections.

Team member	Contribution to this proposal
Zhiyu Song	**TODO: add actual contribution.**
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Zheng Jian	**TODO: add actual contribution.**